

Ex.No:1: Simulation of FDMA in MATLAB

Aim:

To simulate a Frequency Division Multiple Access (FDMA) system using MATLAB and analyze its communication performance in terms of channel bandwidth, data rate, and signal-to-noise ratio (SNR).

Objective 1:

To simulate an FDMA system by allocating individual frequency channels to multiple users and determine the channel bandwidth assigned to each user.

Output Parameter: Channel Bandwidth (kHz)

Program (Matlab Code)

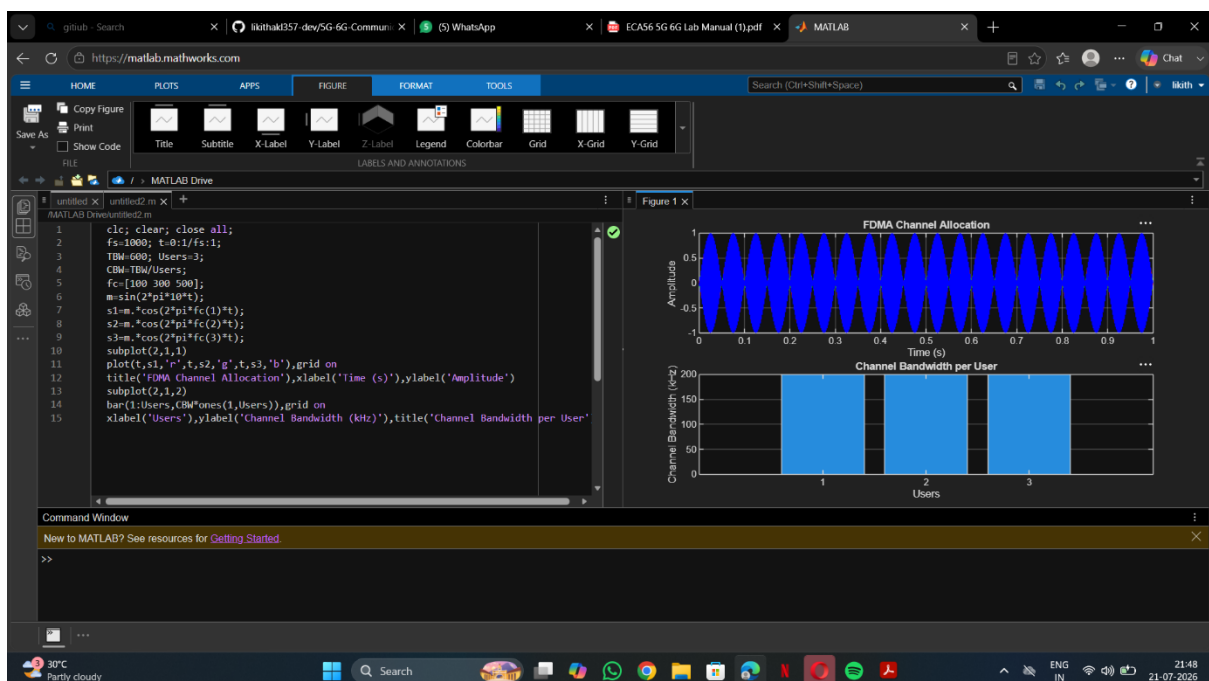
```
clc;
clear;
close all;
% Sampling Parameters
fs = 1000; t = 0:1/fs:1;
% FDMA Parameters
TBW = 600;      % Total Bandwidth (kHz)
Users = 3;      % Number of Users
CBW = TBW / Users; % Channel Bandwidth per User
% Carrier Frequencies
fc = [100 300 500];
% Message Signal
m = sin(2*pi*10*t);
% Modulated Signals
s1 = m .* cos(2*pi*fc(1)*t);
s2 = m .* cos(2*pi*fc(2)*t);
s3 = m .* cos(2*pi*fc(3)*t);
% Plot FDMA Signals
subplot(2,1,1);
plot(t, s1, 'r', t, s2, 'g', t, s3, 'b');
grid on;
title('FDMA Channel Allocation');
```

```

xlabel('Time (s)');
ylabel('Amplitude');
% Plot Channel Bandwidth Allocation
subplot(2,1,2);
bar(1:Users, CBW * ones(1,Users));
grid on;
xlabel('Users');
ylabel('Channel Bandwidth (kHz)');
title('Channel Bandwidth per User');

```

Output Objective 1:



Objective 2:

To calculate the data rate achieved by each user in the FDMA system based on the allocated channel bandwidth.

Output Parameter: Data Rate (kbps)

Objective 2: Matlab Code :

```

clc; clear;

```

```

close all;

% FDMA Parameters

BW = 200; % Channel Bandwidth (kHz)

Users = 3; % Number of Users

M = 2; % BPSK Modulation

% Data Rate Calculation

DataRate = BW * log2(M); % Data Rate (kbps)

Rate = DataRate * ones(1, Users);

% Display Data Rate

disp(' User  Data Rate (kbps)');

disp([(1:Users)' Rate']);

% Plot Data Rate figure;

bar(1:Users, Rate);

xlabel('Users');

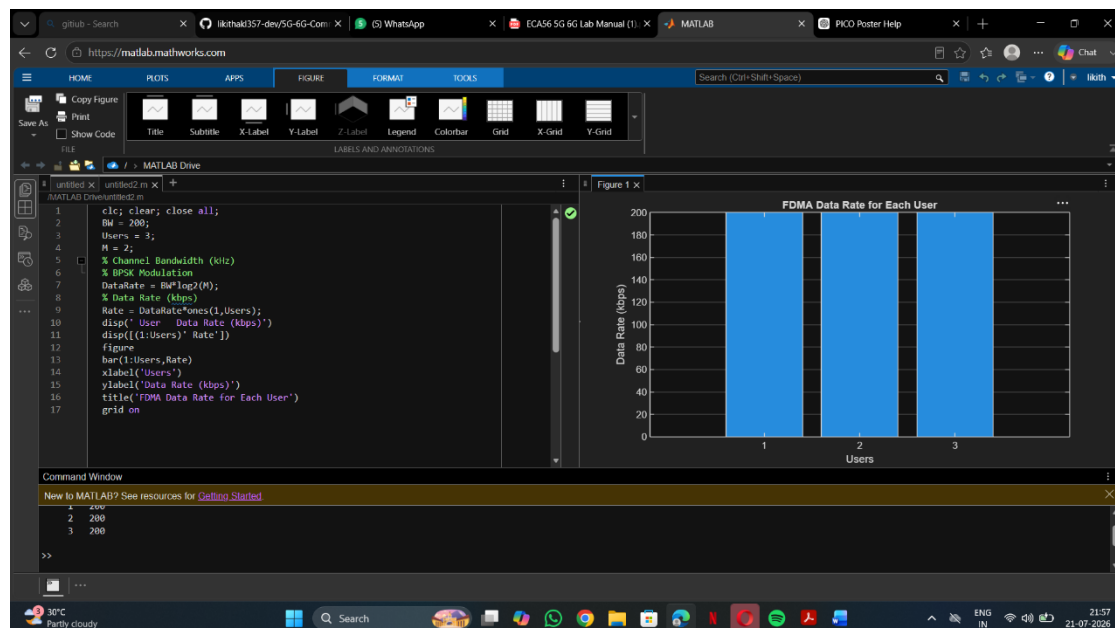
ylabel('Data Rate (kbps)');

title('FDMA Data Rate for Each User');

grid on;

```

Output Objective 2 :



Objective 3:

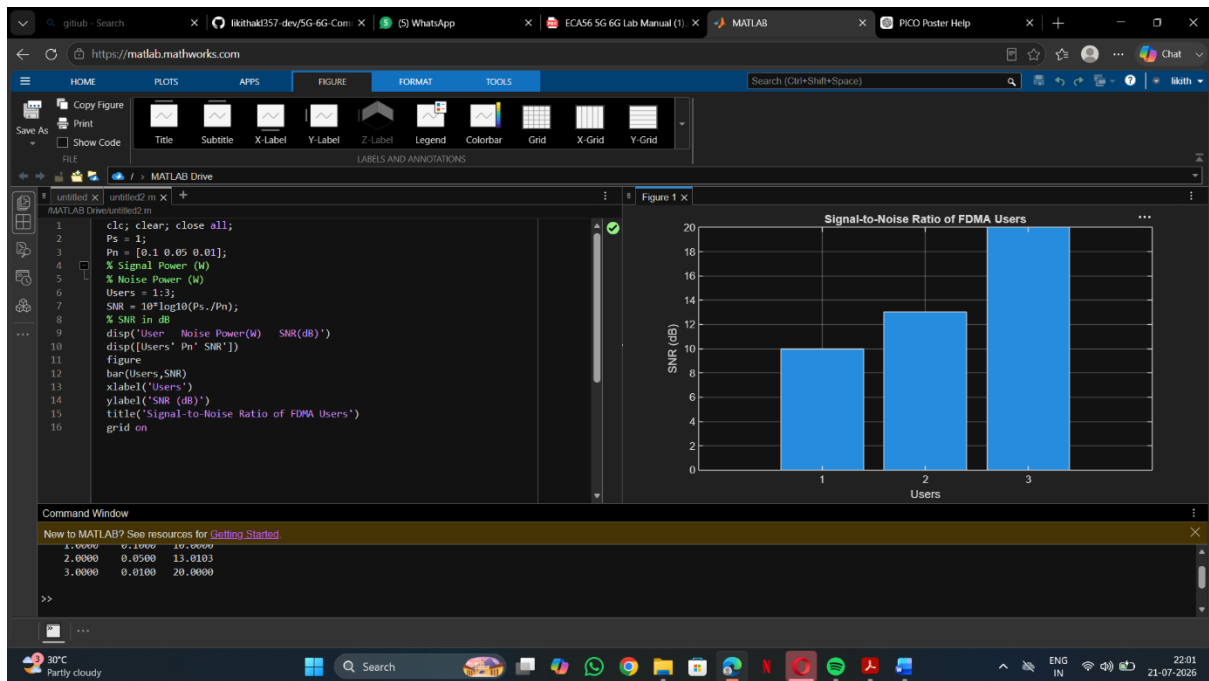
To evaluate the communication quality of the FDMA system by calculating the Signal-to-Noise Ratio (SNR) for different users.

Output Parameter: Signal-to-Noise Ratio (SNR) (dB)

Objective 3: Matlab Code:

```
clc;
clear;
close all;
% Signal and Noise Parameters
Ps = 1; % Signal Power (W)
Pn = [0.1 0.05 0.01]; % Noise Power (W)
% User Information Users = 1:3;
% Calculate Signal-to-Noise Ratio (SNR)
SNR = 10 * log10(Ps ./ Pn); % SNR in dB
% Display Results
disp('User  Noise Power (W)  SNR (dB)');
disp([Users' Pn' SNR']);
% Plot SNR for Each User
figure;
bar(Users, SNR);
xlabel('Users');
ylabel('SNR (dB)');
title('Signal-to-Noise Ratio of FDMA Users');
grid on;
```

Output Objective 3 :



Result:

1. The FDMA system was successfully simulated by allocating separate frequency channels to three users, and the channel bandwidth assigned to each user was determined
2. The data rate achieved by each user was successfully calculated based on the allocated channel bandwidth, demonstrating equal transmission capacity for all users.
3. The Signal-to-Noise Ratio (SNR) for different users was successfully evaluated, showing that higher SNR values provide better communication quality and reliable data transmission.